

Lecture 5: Spherical curves and the fundamental theorem

Recall, a unit-speed curve in $\mathbb{R}^3$ is a **Frenet curve** if $\kappa > 0$. Its positively oriented orthonormal **Frenet frame** is

$$T = c', \quad N = \frac{T'}{\kappa}, \quad B = T \times N.$$

The **torsion** τ is defined by $B' = -\tau N$. Differentiating the inner products of the frame vectors gives the **Frenet equations**:

$$\begin{aligned} T' &= \kappa N, \\ N' &= -\kappa T + \tau B, \\ B' &= -\tau N. \end{aligned}$$

Curvature measures how fast the tangent turns; torsion measures how fast the osculating plane $\text{span}(T, N)$ turns. Curvature is positive here, whereas torsion may have either sign or vanish.

In particular, $\tau \equiv 0$ if and only if the curve is planar: then B is constant and $\langle c, B \rangle$ is constant. Conversely, the binormal of a planar Frenet curve is constant.

Definition. A curve is *spherical* if there are a fixed point m and a constant $R > 0$ such that

$$\|c(s) - m\|^2 = R^2$$

for every s . The point m is the center of the sphere.

Suppose for the moment that the center is known. Differentiating the sphere equation gives

$$\langle m - c, T \rangle = 0.$$

Thus the vector from the curve to the center lies in the normal plane:

$$m - c = aN + bB.$$

Differentiate $\langle m - c, T \rangle = 0$ once more:

$$-1 + \kappa \langle m - c, N \rangle = 0.$$

Consequently $a = 1/\kappa$. Write

$$\rho = \frac{1}{\kappa}, \quad m = c + \rho N + bB.$$

The function ρ is the *radius of curvature*.

The center m must be fixed. The Frenet equations give

$$(c + \rho N + bB)' = (1 - \rho\kappa)T + (\rho' - b\tau)N + (\rho\tau + b')B.$$

Since $\rho\kappa = 1$, the tangent component vanishes automatically. The remaining two equations are

$$\rho' = b\tau, \quad b' = -\rho\tau.$$

If $\tau \neq 0$, the first equation forces

$$b = \frac{\rho'}{\tau}.$$

Substituting this into the second equation gives a condition involving only curvature and torsion:

$$\left(\frac{\rho'}{\tau}\right)' + \rho\tau = 0.$$

This is the spherical-curve criterion. It arises by requiring the candidate center to stop moving.

Theorem (spherical-curve criterion; Kühnel 2.10(ii)). Let $c : I \rightarrow \mathbb{R}^3$ be a unit-speed Frenet curve with $\tau(s) \neq 0$ on I , and put $\rho = 1/\kappa$. Then c is spherical if and only if

$$\left(\frac{\rho'}{\tau}\right)' + \rho\tau = 0.$$

Equivalently, in the notation of the book,

$$\frac{\tau}{\kappa} = \left(\frac{\kappa'}{\tau\kappa^2}\right)'.$$

When this holds, the center and radius are

$$m = c + \rho N + \frac{\rho'}{\tau} B, \quad R^2 = \rho^2 + \left(\frac{\rho'}{\tau}\right)^2.$$

Proof. Necessity was derived on the preceding pages. For sufficiency, define $m(s)$ by the displayed formula. Our calculation gives

$$m'(s) = \left[\rho\tau + \left(\frac{\rho'}{\tau}\right)' \right] B = 0.$$

Thus m is constant. Also $m - c \perp T$, so $(\|m - c\|^2)' = -2\langle m - c, T \rangle = 0$. The distance to m is a positive constant, as required.

There is a local meaning for the same candidate center, even when the curve is not spherical.

Definition. A sphere has contact through third order with c at s_0 if, for its center m and radius R , the function

$$f(s) = \|c(s) - m\|^2 - R^2$$

satisfies $f(s_0) = f'(s_0) = f''(s_0) = f'''(s_0) = 0$. For a smooth curve this says $f(s_0 + h) = O(h^4)$.

Assume $\tau(s_0) \neq 0$, and write $m - c(s_0) = aT + bN + dB$, with the frame evaluated at s_0 . Using

$$c'' = \kappa N, \quad c''' = -\kappa^2 T + \kappa' N + \kappa \tau B,$$

the three derivative conditions give, successively,

$$a = 0, \quad b = \frac{1}{\kappa}, \quad d = -\frac{\kappa'}{\tau\kappa^2} = \frac{\rho'}{\tau}.$$

For example, $f''' = -2\langle m - c, c''' \rangle$, since the curve has unit speed.

This unique sphere is the **osculating sphere** (Kühnel 2.10(i)). For a spherical curve with $\tau \neq 0$, this sphere is the same at every point.

The quotient ρ'/τ cannot be used at a zero of torsion. The two equations from which it came still make sense.

Criterion allowing zero torsion. A unit-speed Frenet curve is spherical if and only if there is a smooth function b on I satisfying

$$\rho' = b\tau, \quad b' = -\rho\tau.$$

Indeed, the same calculation shows that $m = c + \rho N + bB$ is constant, and

$$(\rho^2 + b^2)' = 2\rho b\tau - 2b\rho\tau = 0.$$

Conversely, a sphere center supplies $b = \langle m - c, B \rangle$.

If $\tau \equiv 0$, the criterion requires ρ to be constant. Such a planar curve is an arc of a circle: $c + \rho N$ is its center in its plane. It lies on many spheres, with centers $c + \rho N + bB$ for constant b .

At an isolated zero of torsion on a spherical curve, $\rho' = 0$ as well. One must use the system above or justify a smooth extension of the quotient; division by zero is not allowed.

The criterion also places useful restrictions on spherical curves. Since

$$R^2 = \rho^2 + b^2 \geq \rho^2,$$

every unit-speed curve on a sphere of radius R satisfies

$$\kappa \geq \frac{1}{R}.$$

This also follows directly by differentiating the sphere equation twice, so it does not require nonzero torsion.

If a spherical Frenet curve has constant curvature, then $\rho' = 0$. On any interval where $\tau \neq 0$, the criterion gives $b = 0$, and then $0 = b' = -\rho\tau$, a contradiction. Thus $\tau \equiv 0$, and the curve is an arc of a circle.

A useful caution. Constancy of

$$\rho^2 + \left(\frac{\rho'}{\tau}\right)^2$$

alone does not imply that the curve is spherical. A circular helix with constant $\kappa > 0$ and constant $\tau \neq 0$ makes this expression constant but fails the differential criterion. The center must be fixed, not just the candidate radius.

For a curve on the unit sphere centered at the origin, there is another useful description (Kühnel 2.10(iii)). Suppose

$$\|c\| = 1, \quad \|c'\| = 1, \quad U = c \times T.$$

Differentiating $\langle c, c \rangle = 1$ shows that (c, T, U) is a positively oriented orthonormal frame. Define

$$J = \det(c, c', c'') = \langle c'', U \rangle.$$

Since $\langle c'', c \rangle = -1$ and $\langle c'', T \rangle = 0$, we obtain

$$T' = c'' = -c + JU, \quad U' = -JT.$$

In particular,

$$\kappa^2 = 1 + J^2.$$

The component $-c$ points toward the sphere's center. The component JU is tangent to the sphere and perpendicular to the curve. The signed quantity J is called its **geodesic curvature** on the oriented unit sphere.

The function J also determines torsion. Differentiating $c'' = -c + JU$ gives

$$c''' = -(1 + J^2)T + J'U.$$

Because (c, T, U) is positively oriented,

$$\det(T, -c, U) = 1.$$

The torsion formula for a unit-speed curve now yields

$$\tau = \frac{\det(c', c'', c''')}{\|c' \times c''\|^2} = \frac{J'}{1 + J^2}.$$

Thus the spherical identities are

$$\kappa = \sqrt{1 + J^2}, \quad \tau = \frac{J'}{1 + J^2}.$$

If $J = 0$, then $U' = 0$ and $\langle c, U \rangle = 0$; the curve lies on a great circle. If J is a nonzero constant, the vector $Jc + U$ is constant, and $\langle c, Jc + U \rangle = J$. This is the equation of a plane cutting the sphere in a smaller circle.

Conversely, an oriented circle on the unit sphere has constant J : its torsion is zero, so the displayed formula gives $J' = 0$.

Another relation between curvature and torsion characterizes **slope lines**: Frenet curves whose tangents make a constant angle with a fixed unit vector a .

Theorem (Lancret; Kühnel 2.11). A Frenet curve in $\mathbb{R}^3$ is a slope line if and only if τ/κ is constant.

Proof. If $\langle T, a \rangle$ is constant, then $0 = \kappa \langle N, a \rangle$. Hence

$$a = uT + vB.$$

Here $u = \langle T, a \rangle$ is constant and $v' = \langle B', a \rangle = 0$. Also $v \neq 0$, since otherwise T would be constant. Differentiation gives

$$0 = a' = (u\kappa - v\tau)N, \quad \frac{\tau}{\kappa} = \frac{u}{v}.$$

Conversely, if $\lambda = \tau/\kappa$ is constant, then

$$(\lambda T + B)' = (\lambda\kappa - \tau)N = 0.$$

Thus $a = (\lambda T + B)/\sqrt{1 + \lambda^2}$ is a fixed unit vector, and $\langle T, a \rangle$ is constant.

Example (constant curvature and torsion). Given constants $\kappa_0 > 0$ and $\tau_0 \in \mathbb{R}$, put $\omega = \sqrt{\kappa_0^2 + \tau_0^2}$. Consider

$$c(s) = \left(\frac{\kappa_0}{\omega^2} \cos(\omega s), \frac{\kappa_0}{\omega^2} \sin(\omega s), \frac{\tau_0}{\omega} s \right).$$

Then

$$\|c'\|^2 = \frac{\kappa_0^2 + \tau_0^2}{\omega^2} = 1, \quad \|c''\| = \kappa_0.$$

Taking $N = (-\cos(\omega s), -\sin(\omega s), 0)$ gives

$$B = \left(\frac{\tau_0}{\omega} \sin(\omega s), -\frac{\tau_0}{\omega} \cos(\omega s), \frac{\kappa_0}{\omega} \right), \quad B' = -\tau_0 N.$$

This is a circular helix, or a circle when $\tau_0 = 0$.

The **Darboux vector** $D = \tau T + \kappa B$ satisfies $T' = D \times T$, $N' = D \times N$, and $B' = D \times B$. For constant κ, τ , it is a fixed rotation axis.

We have produced a curve with the prescribed constants. Why must every other curve with these constants have the same shape?

Theorem (fundamental theorem in $\mathbb{R}^3$). Let I be an open interval and let

$$\kappa : I \rightarrow (0, \infty), \quad \tau : I \rightarrow \mathbb{R}$$

be smooth functions. Choose $s_0 \in I$, a point $p_0 \in \mathbb{R}^3$, and a positively oriented orthonormal frame (T_0, N_0, B_0) .

There is a unique smooth unit-speed Frenet curve $c : I \rightarrow \mathbb{R}^3$ with curvature κ , torsion τ , and

$$c(s_0) = p_0, \quad (T, N, B)(s_0) = (T_0, N_0, B_0).$$

Consequently, curvature and torsion, as functions of the same arclength parameter, determine a curve up to a rotation and a translation preserving the orientation of $\mathbb{R}^3$.

Existence means there is no further compatibility equation between κ and τ . The spherical-curve equation is an extra condition characterizing curves on spheres.

The proof reconstructs the frame from its differential equations and then integrates T to reconstruct the curve.

We use one fact about ordinary differential equations.

Linear ODE lemma. If $A(s)$ is a continuous matrix on an interval I , then the initial-value problem

$$X' = A(s)X, \quad X(s_0) = X_0$$

has a unique solution on I . Smooth coefficients give a smooth solution. The unknown may be a vector or a matrix.

Reason. The equation is equivalent to

$$X(s) = X_0 + \int_{s_0}^s A(u)X(u) \, du.$$

On a sufficiently short closed interval about s_0 , this integral operator is a contraction in the supremum norm: its Lipschitz constant is at most the interval's length times $\sup \|A\|$. Successive substitution therefore converges to a unique solution.

The matrix A is bounded on each compact subinterval of I . Repeating the construction on finitely many short intervals extends the solution across that subinterval. Uniqueness makes the extensions agree. This gives a solution on all of I . Differentiating the equation gives the asserted smoothness.

Reconstructing the frame. Put the frame vectors in the rows of a matrix:

$$F = \begin{pmatrix} T^\top \\ N^\top \\ B^\top \end{pmatrix}, \quad K = \begin{pmatrix} 0 & \kappa & 0 \\ -\kappa & 0 & \tau \\ 0 & -\tau & 0 \end{pmatrix}.$$

The Frenet equations are precisely $F' = KF$. The ODE lemma gives a unique solution with the prescribed initial frame. We must check that its rows remain an orthonormal frame.

Since $K^\top = -K$,

$$(F^\top F)' = F^\top (K^\top + K)F = 0.$$

Initially $F^\top F = I_3$, so this remains true. As F is square, also $FF^\top = I_3$; its rows are orthonormal.

An orthogonal matrix has determinant ± 1 . By continuity and the initial condition, $\det F = 1$ throughout I . Thus the frame remains positively oriented, and $B = T \times N$.

Reconstructing the curve. Let T be the first row vector obtained from F , and define

$$c(s) = p_0 + \int_{s_0}^s T(u) \, du.$$

Then $c' = T$, so $\|c'\| = 1$ and $c(s_0) = p_0$. Moreover,

$$c'' = T' = \kappa N.$$

Since $\kappa > 0$ and $\|N\| = 1$, the curvature of c really is the prescribed κ , and its principal normal really is the reconstructed vector N .

Positive orientation gives its binormal $B = T \times N$, and

$$B' = -\tau N$$

shows that its torsion is the prescribed τ .

This verifies existence: solving a matrix equation has produced the actual Frenet frame of a unit-speed curve, not merely three vector fields along an unrelated path.

Uniqueness and rigid motions. Two curves with the same κ, τ and the same initial frame have frame matrices satisfying the same linear initial-value problem. ODE uniqueness makes their frames identical. Their tangent vectors are therefore equal, and equality at s_0 makes the curves equal.

A ***proper Euclidean motion*** is a map

$$x \longmapsto Qx + a, \quad Q^T Q = I_3, \quad \det Q = 1.$$

For $\tilde{c} = Qc + a$, the Frenet frame is (QT, QN, QB) ; its Frenet equations have the same κ, τ . Here proper means that the motion preserves ambient orientation.

Given two curves with the same κ, τ , choose the unique rotation matching their initial frames and then the translation matching their initial points. The preceding uniqueness argument gives $\tilde{c}(s) = Qc(s) + a$ on I .

A reflection preserves κ but changes the sign of τ . This is why the orientation condition matters. In particular, the helix constructed earlier classifies all curves with the same constant curvature and torsion.

The familiar plane reconstruction is the two-dimensional version of the same proof.

For a unit-speed curve in the oriented plane, write

$$T = (\cos \theta, \sin \theta), \quad \nu = (-\sin \theta, \cos \theta).$$

Its ***signed curvature*** k is defined by $T' = k\nu$. Thus $\theta' = k$, and given k we reconstruct by

$$\theta(s) = \theta_0 + \int_{s_0}^s k(u) du,$$

$$c(s) = p_0 + \int_{s_0}^s (\cos \theta(u), \sin \theta(u)) du.$$

The initial angle rotates the curve; the initial point translates it. The ordinary nonnegative curvature is $\kappa = |k|$.

For example, $k(s) = s^{-1/2}$ on $s > 0$ gives $\theta(s) = 2\sqrt{s} + C$, exactly as in Problem 7.

In space, a single angle no longer describes the frame. The matrix equation $F' = KF$ takes the place of $\theta' = k$; the last step, integrating the tangent vector, is the same.

We now formulate the general theorem in $\mathbb{R}^n$, with its standard orientation and $n \geq 2$.

Definition. A smooth unit-speed curve $c : I \rightarrow \mathbb{R}^n$ is a **Frenet curve** if

$$c', c'', \dots, c^{(n-1)}$$

are linearly independent at every point. Here $c^{(j)}$ means the j -th derivative with respect to s .

Apply Gram–Schmidt to this ordered list. Explicitly,

$$v_i = c^{(i)} - \sum_{j < i} \langle c^{(i)}, e_j \rangle e_j, \quad e_i = \frac{v_i}{\|v_i\|} \quad (1 \leq i \leq n-1).$$

The independence assumption ensures $v_i \neq 0$. Choose the last unit vector e_n perpendicular to the others so that $(e_1, \dots, e_n)$ is positively oriented. This is the **Frenet n -frame**.

For $n = 3$ it is (T, N, B) . For $n = 2$, every regular unit-speed curve qualifies, and the frame is (T, ν) , including points where signed curvature vanishes.

Theorem (Frenet equations; Kühnel 2.13). There are functions $\kappa_1, \dots, \kappa_{n-1}$ such that

$$\begin{aligned} e'_1 &= \kappa_1 e_2, \\ e'_i &= -\kappa_{i-1} e_{i-1} + \kappa_i e_{i+1} \quad (2 \leq i \leq n-1), \\ e'_n &= -\kappa_{n-1} e_{n-1}. \end{aligned}$$

The first $n-2$ curvatures are positive. The last curvature may have either sign or vanish.

Why only neighboring frame vectors occur. For $i < n$, the vector e_i lies in $\text{span}(c', \dots, c^{(i)})$, so e'_i lies in $\text{span}(e_1, \dots, e_{i+1})$. Also orthonormality gives

$$\langle e'_i, e_j \rangle = -\langle e_i, e'_j \rangle.$$

These two facts eliminate every coefficient except the two neighbors. Set $\kappa_i = \langle e'_i, e_{i+1} \rangle$. For $i \leq n-2$, Gram-Schmidt gives $\kappa_i = \|v_{i+1}\|/\|v_i\| > 0$.

For $n=3$, $(\kappa_1, \kappa_2) = (\kappa, \tau)$. For $n=2$, the single curvature is $\kappa_1 = k$, the *signed* plane curvature; the positivity requirement is empty.

Theorem (fundamental theorem; Kühnel 2.15). Let $\kappa_1, \dots, \kappa_{n-1}$ be smooth functions on an open interval I , with

$$\kappa_1, \dots, \kappa_{n-2} > 0.$$

Fix $s_0 \in I$, a point $p_0 \in \mathbb{R}^n$, and a positively oriented orthonormal frame $(e_1^0, \dots, e_n^0)$.

There is exactly one smooth unit-speed Frenet curve with these Frenet curvatures and initial data

$$c(s_0) = p_0, \quad e_i(s_0) = e_i^0 \quad (1 \leq i \leq n).$$

Consequently, the Frenet curvatures determine the curve up to a proper Euclidean motion.

Existence of the frame. Let F have rows $e_i^\top$. Define the matrix K by

$$K_{i,i+1} = \kappa_i, \quad K_{i+1,i} = -\kappa_i,$$

with all other entries zero. Solve $F' = KF$ with the given initial frame. The ODE lemma and $K^\top = -K$ imply, just as in dimension three, that F stays orthogonal with determinant 1. Define $c(s) = p_0 + \int_{s_0}^s e_1(u) du$.

We must still verify the Frenet independence condition for the reconstructed curve.

Successive differentiation of $c' = e_1$, using the Frenet equations, gives

$$c^{(j)} = \sum_{i < j} a_{ji} e_i + (\kappa_1 \cdots \kappa_{j-1}) e_j, \quad 1 \leq j \leq n-1.$$

The empty product for $j = 1$ is 1. To see the induction, differentiate the leading term: only its derivative contributes an e_{j+1} -component, and that coefficient is multiplied by κ_j .

Every leading coefficient in this triangular system is positive. Therefore $c', \dots, c^{(n-1)}$ are independent, and their Gram–Schmidt frame is exactly $e_1, \dots, e_{n-1}$. Orientation determines e_n . Thus the reconstructed curve has precisely the prescribed Frenet curvatures.

Uniqueness. The matrix ODE determines the frame uniquely; integrating $c' = e_1$ with the given initial point determines the curve uniquely. Matching initial frames and points by a proper Euclidean motion proves the final assertion of the theorem.

Consequence. A Frenet curve in $\mathbb{R}^n$ lies in an affine hyperplane if and only if $\kappa_{n-1} \equiv 0$.

If $\kappa_{n-1} = 0$, the last Frenet equation makes e_n constant, and

$$(\langle c, e_n \rangle)' = \langle e_1, e_n \rangle = 0.$$

Thus $\langle c, e_n \rangle$ is a fixed affine hyperplane equation. Conversely, if c lies in a hyperplane, its first $n - 1$ derivatives span the direction space of that hyperplane. Hence e_n is its constant unit normal, and the last Frenet equation forces $\kappa_{n-1} = 0$.

Constant prescribed curvatures (Kühnel 2.16). When K is constant, the frame has the explicit solution

$$F(s) = \exp((s - s_0)K)F(s_0), \quad \exp(uK) = \sum_{j=0}^{\infty} \frac{u^j K^j}{j!}.$$

Differentiating the convergent series verifies the equation and initial value. Integrating its first row gives the curve. For $n = 3$, this recovers circles and circular helices.

The spherical criterion and the fundamental theorem can now be used together to construct examples.

Choose $R > 0$, $a \neq 0$, and the interval $|as| < \pi/2$. Prescribe

$$\rho(s) = R \cos(as), \quad \kappa(s) = \frac{1}{R \cos(as)}, \quad \tau(s) = a.$$

Then

$$b(s) = \frac{\rho'}{\tau} = -R \sin(as), \quad b' = -aR \cos(as) = -\rho\tau.$$

The fundamental theorem produces a unit-speed curve with these κ, τ . The spherical criterion then proves that its image lies on a sphere, with

$$\rho^2 + b^2 = R^2.$$

For $c(0) = 0$ and the standard initial frame,

$$m = c(0) + \rho(0)N(0) + b(0)B(0) = (0, R, 0).$$

This gives a nonplanar spherical curve without explicitly solving for its coordinates. Existence comes from the fundamental theorem; the geometry of its image comes from the differential relation between curvature and torsion.